

Your project proposal should be **5-8 pages** (excluding references) and should include the following components (you should already have good starts on some of these from your proposal submission):

- Title
- Abstract: a short summary that briefly answers the following 4 questions
 - What is the problem?
 - Why is the problem of interest?
 - What is the proposed solution/hypothesis?
 - What evidence do you have to support your solution/hypothesis?
- Introduction:
 - A clear description of the problem you are investigating, both abstractly and in context of a particular experimental domain
 - A brief justification for why this problem is interesting and relevant to the class
 - A brief summary of the contributions or interesting findings of your project
- Related Work:
 - References to at least 5 papers that are relevant to the subject of interest.
 - Compare and contrast your approach with the related work.
- Problem Statement
 - Clearly describe the problem or goal of your work. Use mathematical notation if possible to formalize the problem. Make it as general and solution agnostic as possible.
- Method:
 - Clearly state your key insight or idea for addressing your problem
 - Include a high-level and low-level details of your methodology
- Experimental Design:
 - One or more testable hypotheses
 - A concrete domain in which to test each hypothesis along with a description of the domain/data
 - A description of one or more experiments you used to test your hypothesis and a clear evaluation criteria to determine the outcome of your experiment / hypothesis.
- Results and Discussion
 - An analysis of your results
 - Professional looking tables, figures, and plots that present your results
 - Discussion about whether the results confirm your hypothesis
 - Dig into your results and try and figure out what worked and what didn't work and why

- Run and discuss follow-on experiments that dig deeper into your initial results
- Conclusion and Future Work
 - Recap what your problem is and what is interesting about your findings
 - What did you learn or discover?
 - Talk about what future research would look like. For example, if you had another month or two to work on this project what would you do or explore?
- Code:
 - Please include a github link to your code in your writeup for reproducibility.

Submission: Please submit your full project as a PDF on Canvas. Use the Overleaf template that you used for the proposal ([Overleaf template](#) Just go to Menu and select copy project)

Note that you don't have to use overleaf, but you should still download the tex and style files and use the template even if you compile locally.

Make sure to list all group members' names on proposal and only turn in one proposal per group.